Tom George

Post-doctoral fellow in ML and theoretical neuroscience,
also building open-source software for science
Mila | Google | UCL | Harvard | Cambridge

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ACADEMIC POSITIONS

Post-doctoral Fellow and Canadian Neuroanalytic Scholar, Mila | Research on foundation models for neuroscience and NeuroAI with Prof. Blake Richards (McGill/Mila) and Prof. Guillaume Lajoie (UdeM/Mila). 2025–

EDUCATION

PhD, Sainsbury Wellcome Centre, UCL | Advisors: Prof. Caswell Barry (UCL), Prof. Claudia Clopath (Imperial) and Dr. Kimberly Stachenfeld (Google DeepMind/Columbia). Member of **Google DeepMind NeuroLab**. 2020–25

Herchel Smith Scholarship, Harvard University | Prestigious and fully funded scholarship to study at Harvard; 4.0 GPA. Advisors: Prof. Cengiz Pehlevan & Prof. Sam Gershman. 2019–20

BA, MA & MSci, Physics (Natural Sciences), University of Cambridge | 1st Class (Part III), 1st Class (Part II), 1st Class (Part I). 2015–19

AWARDS & FUNDING

Canadian Neuroanalytic Scholarship | Postdoctoral fellowship valued at CAD 140,000. 2026–28

UCL Early Career Neuroscience Prize | Junior category, £500. 2023

Herchel Smith Scholarship | $80,000 fully funded scholarship to Harvard. 2019

Emmanuel College Cambridge John Mainhood Prize | for high academic achievement. 2016–18

SELECTED PUBLICATIONS

T. M. George (2025) | SIMPL: Scalable and hassle-free optimisation of neural representations from behaviour. *International Conference on Learning Representations (**ICLR**)*. Paper | Code.

T. M. George (2023) | A generative model of the hippocampal formation trained with theta driven local learning rules. *Advances in Neural Information Processing Systems (**NeurIPS**)*. Paper.

E. J. Thompson, …, **T. M. George** et al. (2026) | Replay of procedural memory is independent of the hippocampus. ***Nature Neuroscience***. Paper.

T. M. George (2023) | Theta sequences as eligibility traces: A biological solution to credit assignment. ***Tiny Papers Track at ICLR 2023***, Kigali, Rwanda (accepted for oral presentation). Paper.

T. M. George, M. Rastogi, W. de Cothi, C. Clopath, K. Stachenfeld, & C. Barry (2024) | RatInABox: An open source toolkit for modelling locomotion and neuronal activity in continuous environments. ***eLife***. Paper | Code.

T. M. George*, W. de Cothi*, K. Stachenfeld, & C. Barry (2023) | Rapid learning of predictive maps with STDP and theta phase precession. ***eLife***; 70 citations. Paper | Code.

T. M George, G. E. Manucharyan, & A. F. Thompson (2021) | Deep learning to infer eddy heat fluxes from sea surface height patterns of mesoscale turbulence. ***Nature Communications***; 80 citations. Paper.

T. M. George & P. Liò (2019) | Unsupervised Machine Learning for Data Encoding applied to Ovarian Cancer Transcriptomes. ***bioRxiv***. Paper.

SOFTWARE

RatInABox | Creator and lead maintainer of an open-source scientific Python package that generates motion and neural data in spatial environments; 75,000+ PyPI downloads and an *eLife* Tools and Resources article. Code. 2022

SIMPL | Originated and implemented a scalable neural latent-variable method, published at ICLR 2025 with open-source code and data. Code. 2024

INDUSTRY

X AI Residency, Google | Completed a five-month AI residency developing foundation models for geophysics at X, Google's moonshot factory, in Mountain View. 2025

ML consultant, Mltpl | Developed machine-learning tools for automated business valuations. 2023–24

TECHNICAL SKILLS

Programming | Python, JAX, PyTorch, TensorFlow, bash and Julia. GitHub.

Machine Learning | Foundation models, representation learning, latent-variable models, RL, NLP, CNNs and RNNs.

Languages | English (Native), Spanish (DELE B1 equivalent).

TEACHING & OUTREACH

TReND-CaMinA | Co-founder and organizer of a recurring, fully funded computational neuroscience and ML school across Ghana, Rwanda, Zambia and Kenya. Co-developed its open-source curriculum and secured $300,000 in funding. 2023–26

NeuroAI summer school | University of Amsterdam, invited lecturer on model-free RL. 2024

Open-source teaching materials | Created open teaching materials for deep learning, modern Hopfield networks, reinforcement learning, spatial representations, path integration and successor features. Materials.

SELECTED INVITED TALKS

Mila, Quebec AI Institute | SIMPL; fast latent variable analysis for neural data. 2025

Oxford University, Cortex Club | RatInABox (see open source). 2024

Google DeepMind, London | Optimizing internal representations (NeuroLab workshop).

Imperial, Pint of Science | Public outreach talk on "The language of the brain".

ICLR TinyPapers workshop oral | Neural oscillations and eligibility traces, Kigali, Rwanda. 2023
Google DeepMind, London | The Helmholtz hippocampus (NeuroLab workshop).
UCL Neuroscience Symposium (prize winner) | How hippocampus learns predictive maps?
Spring Hippocampal Research Conference, Verona | RatInABox (see open source).
Mila Neural-AI reading group | Biological Reinforcement Learning in the Hippocampus. 2022

ACADEMIC SERVICE

Area chair, ICLR 2023 and 2024 | TinyPapers workshop.
Reviewer, NeurIPS 2023 | Associative Memory & Hopfield Networks workshop.

SELECTED POSTERS

COSYNE 2025 poster | **T. M. George**, P. Glaser, K. Stachenfeld, C. Barry & C. Clopath. Repeated decoding optimises neural representations and is a SIMPL approach to finding latent variables in spiking data. Poster. 2025

COSYNE 2024 poster | C. Barry, M. Rastogi, W. de Cothi, C. Clopath, K. Stachenfeld, **T. M. George** (2024). RatInABox: A unified Python framework for modelling spatial behaviour and neural data.

COSYNE 2024 poster | **T. M. George**, C. Barry, K. Stachenfeld, C. Clopath, T. Fukai (2024). The Helmholtz Hippocampus: A biologically plausible generative model of the hippocampal formation.

NeurIPS 2023 poster | **T. M. George**, C. Barry, K. Stachenfeld, C. Clopath, T. Fukai (2024). The Helmholtz Hippocampus: A biologically plausible generative model of the hippocampal formation.

COSYNE 2024 poster | **T. M. George**, W. de Cothi, K. Stachenfeld, C. Barry (2022). Rapid learning of predictive maps with STDP and theta phase precession.

SFN 2022 poster & Cosyne 2023 talk | E. Thompson, L. Rollik, **T. M. George*** et al. Replay of motor sequences in DLS during consolidation using an unsupervised point process model.

SFN 2021 poster | A. Onih, **T. M. George**, S. Nierwetberg & A. Akrami. Pupil dilation as a proxy for statistical learning in freely moving mice and humans.

EARLIER PLACEMENTS

Caltech | Summer Undergraduate Research Fellow with Prof. Andrew Thompson; the project led to a *Nature Communications* paper on deep-learning prediction of eddy heat fluxes. 2018
Okinawa Computational Neuroscience Course (OCNC) | Summer school participant. 2022
Okinawa Institute of Science and Technology (OIST) | Research intern with Prof. Tomoki Fukai. 2022